An Extended Continuation Power Flow Method for Static Voltage Stability Assessment of Renewable Power Generation-Penetrated Power Systems

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***Abstract*—The continuation power flow method (CPF) has been extensively used for obtaining the static voltage stability limit of power systems, which provides important guidance for the stable operation. However, the conventional CPF cannot account for the dynamic characteristics of renewable power generation, thus resulting in prominent errors for renewable power generation-penetrated power systems. In this brief, we investigate the static voltage stability assessment (SVSA) of power systems with high penetration of renewable power generation (RPG). An extended Jacobian matrix is firstly established by considering the dynamic characteristics of synchronous generators (SG) and renewable power generation in constant AC voltage control mode (CVRPG). Based on the extended Jacobian matrix, an extended CPF method is proposed for SVSA of the power system and a reactive power sensitivity index is proposed to indicate the voltage support capability of nodes. Simulation results show that the proposed extended CPF method has higher accuracy than conventional CPF in assessing the static voltage stability of power systems with high penetration of RPG. It is also revealed that the voltage support capability of each node rises as the proportion coefficient is increases.**

***Index Terms*—Continuation power flow, static voltage stability, renewable power generation, extended Jacobian matrix.**

I. Introduction

STATIC voltage stability characterizes the ability of a power system to retain stable voltages under small disturbances [1], [2]. Assessing the static voltage stability is crucially important for power system operation. Recently, the increasing integration of renewable power generation (RPG) brings great challenges to the static voltage stability assessment (SVSA), due to the distinct dynamic characteristics of RPG compared with conventional power plants.

A set of previous studies investigated the influence of RPG on the power system's static voltage stability. A voltage stability margin index was proposed to reveal the relationship between control parameters and the stability margin in real-time [3]. This index can reflect the influence of different penetration levels of RPG on the static voltage stability limit of the power system but with lower accuracy. The critical short-circuit ratio were proposed in the literature to assess the static voltage stability of the power system with integration of high voltage direct current transmission (HVDC) [4], [5], [6], [7]. A modified short-circuit ratio was further proposed [8] by taking into account the impact of RPG. However, due to the need for simultaneous solution based on detailed model, it is too complicated to be applied to large-scale power systems. Nodal voltage sensitivity that measures the weakness of a node was adapted in [9] to study the impact of RPG on the power system's voltage stability. But the RPG nodes are treated as constant power sources without considering their ability to regulate the AC voltage. Data-driven methods based on AI techniques were proposed in [10], [11], which have the potential for fast SVSA of power systems containing large-scale RPG. But they require a large amount of online data and the reliability of the results cannot be secured. Time-domain simulations were used in [12] for assessing the static voltage stability of power systems penetrated by RPG. This method ensures high accuracy of the analysis results, but takes huge computational costs.

The continuation power flow method (CPF) has also been most commonly used for SVSA of power systems, due to its high analysis accuracy and applicability to large-scale power systems. CPF obtains a continuum of power flow solutions from the intial power to the static voltage stability limit power of power systems, and the criterion of reaching the limit is whether the minimum eigenvalue of Jacobian matrix is zero. However, its applicability declines with the continuous integration of RPG. In the conventional CPF, synchronous generators (SG) are generally assumed with a constant voltage. This treatment introduces insignificant errors because SG has a nearly constant internal electromotive force (EMF) and relatively small transient reactance. But when applied to RPG, this treatment poses prominent errors due to the absence of constant internal EMF [13], [14], [15].

In order to achieve fast and accurate SVSA for power systems with high penetration of RPG, it is necessary to improve the conventional CPF method by considering the dynamic characteristics of RPG. In addition, a voltage support

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ability index of nodes applicable to RPG is also conducive to the rapid SVSA, due to that the reactive power sensitivity can reflect the static voltage stability level [16]. The main contributions of this brief are summarized as follows:

1) An extended Jacobian matrix is established that considers the dynamic characteristics of SG and CVRPG;
2) An extended CPF method based on the extended Jacobian matrix was proposed to realize the accurate SVSA of power systems with high penetration of RPG;
3) A reactive power sensitivity index is proposed to indicate the capability of nodes in supporting static voltage stability of the power system.

The rest of this brief is organized as follows: In Section II, the power flow model is introduced where the extended Jacobian matrix is established by considering the dynamics of SG and CVRPG nodes. In Section III, the extended CPF method is proposed for SVSA of power systems with high penetration of RPG and reactive power sensitivity index is proposed to measure the voltage support capability of nodes. In Section IV, simulations are performed to verify the accuracy of the proposed SVSA method and to examine the influence of control parameters.

II. Power Flow Model and Extended Jacobian Matrix

A. Steady-State Power Flow Model

For a system with m nodes, the power flow equations of each node are:

$$\begin{cases} P_i = P_{Gi} - P_{Li} = U_i \sum_{j\in i} U_j\left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) \\ Q_i = Q_{Gi} - Q_{Li} = U_i \sum_{j\in i} U_j\left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) \end{cases} \tag{1}$$

where $i = 1, 2, \ldots, m$; $j \in i$ represents that node j connects to node i directly; P_{Gi} and Q_{Gi} are the active power and reactive power injected from the generators of node i; P_{Li} and Q_{Li} are the active power and reactive power load of node i; G_{ij} and B_{ij} are the real part and imaginary part of the admittance of transmission line (i, j); θ_{ij} is the phase angle difference between node i and node j.

Set the node m as the slack node and let $n = m-1$. r is the number of PV nodes. The variables to be solved are the phase angles of the PQ and PV nodes, and the voltage magnitudes of the PQ nodes. Equations (1) can be rewritten as

$$\begin{cases} \Delta P_i = P_{Gi} - P_{Li} - U_i \sum_{j\in i} U_j\left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) \\ \Delta Q_i = Q_{Gi} - Q_{Li} - U_i \sum_{j\in i} U_j\left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) \end{cases} \tag{2}$$

With the *Newton-Raphson* method, equations (2) can be further written as

$$\begin{bmatrix} \Delta \boldsymbol{P} \\ \Delta \boldsymbol{Q} \end{bmatrix} = \boldsymbol{J} \begin{bmatrix} \Delta \boldsymbol{\theta} \\ \Delta \boldsymbol{U} \end{bmatrix} \begin{matrix} n \\ n-r \end{matrix} \tag{3}$$

where $\boldsymbol{J}$ is the Jacobian matrix:

$$\boldsymbol{J} = \begin{bmatrix} \dfrac{\partial \Delta \boldsymbol{P}}{\partial \boldsymbol{\theta}} & \dfrac{\partial \Delta \boldsymbol{P}}{\partial \boldsymbol{U}} \\ \dfrac{\partial \Delta \boldsymbol{Q}}{\partial \boldsymbol{\theta}} & \dfrac{\partial \Delta \boldsymbol{Q}}{\partial \boldsymbol{U}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{H} & \boldsymbol{N} \\ \boldsymbol{M} & \boldsymbol{L} \end{bmatrix} \begin{matrix} n \\ n-r \end{matrix} \quad \begin{matrix} n & n-r \end{matrix} \tag{4}$$

P_{Gi}, P_{Li} and Q_{Li} are considered to be constants in steady-state power flow model. Thus, the off-diagonal elements H_{ij}, N_{ij}, M_{ij}, L_{ij} $(i \neq j)$ of the blocks $\boldsymbol{H}$, $\boldsymbol{N}$, $\boldsymbol{M}$ and $\boldsymbol{L}$ in $\boldsymbol{J}$ can be calculated with the following formulas

$$\begin{cases} H_{ij} = \dfrac{\partial \Delta P_i}{\partial \theta_j} = -U_i U_j\left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) \\ N_{ij} = \dfrac{\partial \Delta P_i}{\partial U_j} = -U_i\left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) \\ M_{ij} = \dfrac{\partial \Delta Q_i}{\partial \theta_j} = U_i U_j\left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) \\ L_{ij} = \dfrac{\partial \Delta Q_i}{\partial U_j} = -U_i\left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) \end{cases} \tag{5}$$

The diagonal elements H_{ii}, N_{ii}, M_{ii}, L_{ii} can be calculated with

$$\begin{cases} H_{ii} = \dfrac{\partial \Delta P_i}{\partial \theta_i} = U_i \sum_{j\in i, j\neq i} U_j\left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) \\ N_{ii} = \dfrac{\partial \Delta P_i}{\partial U_i} = -\sum_{j\in i, j\neq i} U_j\left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) - 2U_i G_{ii} \\ M_{ii} = \dfrac{\partial \Delta Q_i}{\partial \theta_i} = -U_i \sum_{j\in i, j\neq i} U_j\left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) \\ L_{ii} = \dfrac{\partial \Delta Q_i}{\partial U_i} = -\sum_{j\in i, j\neq i} U_j\left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) + 2U_i B_{ii} \end{cases} \tag{6}$$

B. Extended Jacobian Matrix by Considering Nodal Dynamics

In the steady-state power flow model, generator nodes are described as PV nodes whose output power and terminal voltage magnitude are assumed to be constant. This assumption eliminates the $r+1$ equations related to the reactive power/voltage of the generator node and the derivatives of P_{Gi} and Q_{Gi} with respect to U_i and θ_i in H_{ii} and N_{ii}. In order to make the analysis more accurate for power systems with high penetration of RPG, the assumption needs to be modified by considering nodal dynamics.

1) Dynamics of SG Nodes: For SGs, the assumption that the internal EMF E' is constant is more accurate, under which the SG node is described as a PQ node and the terminal magnitude and phase angle become the variables to be solved. The active power and reactive power of the SG node are shown as,

$$\begin{cases} P_{Gi} = \dfrac{E'_i U_i \sin\theta_{\delta i}}{x'_{di}} \\ Q_{Gi} = \dfrac{\left(E'_i U_i \cos\theta_{\delta i} - U_i^2\right)}{x'_{di}} \end{cases} \tag{7}$$

Their differential with respect to U_i and θ_i are shown as

$$\begin{cases} \dfrac{\partial P_{Gi}}{\partial \theta_i} = -\dfrac{E'_i U_i \cos\theta_{\delta i}}{x'_{di}} & \dfrac{\partial P_{Gi}}{\partial U_i} = \dfrac{E'_i \sin\theta_{\delta i}}{x'_{di}} \\ \dfrac{\partial Q_{Gi}}{\partial \theta_i} = \dfrac{E'_i U_i \sin\theta_{\delta i}}{x'_{di}} & \dfrac{\partial Q_{Gi}}{\partial U_i} = \dfrac{\left(E'_i \cos\theta_{\delta i} - 2U_i\right)}{x'_{di}} \end{cases} \tag{8}$$

The diagonal elements of the blocks $\boldsymbol{H}$, $\boldsymbol{N}$, $\boldsymbol{M}$ and $\boldsymbol{L}$ in Jacobian matrix corresponding to SG nodes are modified from equation (6) to

$$\begin{cases} H'_{ii} = U_i \sum_{j\in i, j\neq i} U_j\left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) + \dfrac{\partial P_{Gi}}{\partial \theta_i} \\ N'_{ii} = -\sum_{j\in i, j\neq i} U_j\left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) - 2U_i G_{ii} \dfrac{\partial P_{Gi}}{\partial U_i} \\ M'_{ii} = -U_i \sum_{j\in i, j\neq i} U_j\left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) + \dfrac{\partial Q_{Gi}}{\partial \theta_i} \\ L'_{ii} = -\sum_{j\in i, j\neq i} U_j\left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) + 2U_i B_{ii} + \dfrac{\partial Q_{Gi}}{\partial U_i} \end{cases} \tag{9}$$

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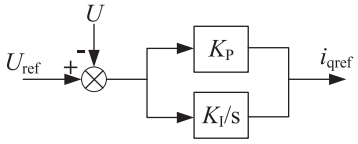


Fig. 1. Typical control diagram of RPG that participates in voltage regulation.

2) Dynamics of CVRPG Nodes: For nodes containing CVRPG, it is also necessary to correct its corresponding Jacobian matrix elements. The typical control diagram of RPG that participates in voltage regulation is shown in Fig. 1.

Since this brief is concerned with the timescale at tens of milliseconds, the following assumptions are made to ensure high accuracy without overcomplicating the analysis:

1) Ignore the dynamics of the phase locking loop of the RPG, that is the phase angle used by the controller is the exact value of the power system;
2) Ignore the dynamic response process of the current controller of the RPG, that is the actual value of the current is equal to the output of the voltage controller;
3) Ignore the dynamic response process of the integrator of the voltage PI controller, since the dynamic response output of the integrator is much smaller than that of the proportional tuning. So output of the controller is the sum of the proportional tuning and the historical value of the integrator i_{q0i}.

Based on the above assumptions, the active and reactive power output by CVRPG can be deduced as

$$\begin{cases} P_{Gi} = U_i i_{di} \\ Q_{Gi} = U_i i_{qi} \\ i_{qi} = i_{q0i} + K_{Pi}\left(U_{refi} - U_i\right) \end{cases} \tag{10}$$

where K_{Pi} and U_{refi} are the proportional coefficient and voltage reference of the ith voltage controller, i_d and i_q are the d-axis and q-axis components of the current, respectively.

Linearizing equation (10) at the steady-state operating point, we can get

$$\begin{cases} \Delta P_{Gi} = \frac{P_{Gi}}{U_i}\Delta U_i \\ \Delta Q_{Gi} = \left(\frac{Q_{Gi}}{U_i} - K_{Pi}U_i\right)\Delta U_i \end{cases} \tag{11}$$

The RPG deploys power electronic devices that has zero short-term overload capability. When its reactive current reaches the upper limit i_{qmax}, it will exit the constant voltage control mode. Then, the differentials of the active and reactive powers with respect to U_i and E_i are

$$\begin{aligned} &\frac{\partial P_{Gi}}{\partial \theta_i} = 0,\ \frac{\partial P_{Gi}}{\partial U_i} = \frac{P_{Gi}}{U_i},\ \frac{\partial Q_{Gi}}{\partial \theta_i} = 0 \\ &\frac{\partial Q_{Gi}}{\partial U_i} = \begin{cases} \frac{Q_{Gi}}{U_i} - K_{Pi}U_i, & i_{qi} < i_{q\max i} \\ 0, & i_{qi} \geq i_{q\max i} \end{cases} \end{aligned} \tag{12}$$

By substituting equation (12) into equation (9), the corresponding elements in Jacobian matrix are modified.

3) Extended Jacobian Matrix: With the above considerations of both SG and CVRPG nodes, $r+2$ equations are added to equations (2) and (3) respectively, the total number of equations will be $2m$. The Jacobian matrix is enlarged to a $2m \times 2m$ dimensional matrix and is called the extended Jacobian matrix $\boldsymbol{J}'$. Thus, the power flow model considering nodal dynamics can be obtained as

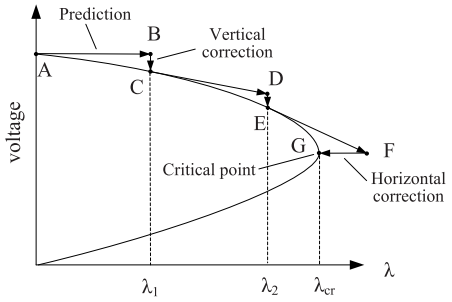


Fig. 2. Traditional continuation power flow calculation process.

$$\begin{bmatrix} \Delta \boldsymbol{P} \\ \Delta \boldsymbol{Q} \end{bmatrix}_m = \boldsymbol{J}' \begin{bmatrix} \Delta \boldsymbol{\theta} \\ \Delta \boldsymbol{U} \end{bmatrix}_m \tag{13}$$

The extended Jacobian matrix is shown as follows,

$$\boldsymbol{J}' = \begin{bmatrix} \boldsymbol{H}' & \boldsymbol{N}' \\ \boldsymbol{M}' & \boldsymbol{L}' \end{bmatrix}_m^m \tag{14}$$

where the diagonal elements of $\boldsymbol{H}'$, $\boldsymbol{N}'$, $\boldsymbol{M}'$ and $\boldsymbol{L}'$ are shown in equation (9), and the off-diagonal elements are shown in equation (5).

III. Static Voltage Stability Assessment With Extended CPF

A. Extended CPF Based on Extended Jacobian Matrix

The basic process of the conventional CPF to obtain the static voltage stability limit can be briefly summarized as: Firstly, the ratio of the current load power to the rated load power is used to the indicate the loading level, represented as λ. Then, continuously increase λ and do the power flow calculation. When λ exceeds a specific limit, the determinant of the Jacobian matrix changes sign, which indicates the static voltage instability. As the determinant sign is the same with the sign of the minimum eigenvalue, that the minimum eigenvalue is zero or close to zero is generally set as the criterion of CPF to obtain the static voltage stability limit [17], [18]. The calculation process of CPF is as shown in Fig. 2.

In order to make conventional CPF more suitable for power systems with high penetration of RPG, two main improvements are made. First, the extended Jacobian matrix $\boldsymbol{J}'$ is used. Second, the current limit of CVRPG nodes is considered in the calculation, i.e., when the calculated current of one CVRPG node reaches its limits, the control mode is changed to constant reactive power control. The detailed calculation process of the extended continuation power flow is as follows:

1) Perform power flow calculation for the initial loading condition to obtain the voltage magnitude, phase angle, and active and reactive power generation and load at each node.
2) Increase λ and perform prediction and correction based on the previous step of power flow calculation results to obtain the updated power flow calculation results.

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3) Update the control mode of CVRPG nodes according to the convergent power flow calculation results.
4) Obtain the extended Jacobian matrix $\boldsymbol{J}'$.
5) Judge whether the minimum eigenvalue of $\boldsymbol{J}'$ is zero or close to zero. If yes, the static voltage stability limit is reached; Otherwise, return to step 2).

By incorporating the dynamics of SG and CVRPG nodes in the conventional power flow model, the extended CPF based on $\boldsymbol{J}'$ makes the solution of the static voltage stability limit more accurate. Meanwhile, $\boldsymbol{J}'$ slightly increases the computation cost by merely introducing several terms in the diagonal elements compared with $\boldsymbol{J}$, and that is only a small part of the whole CPF calculation process.

B. Evaluation of Voltage Support Capability

The voltage sensitivity factor (VSF) has been widely used in the research of voltage stability to indicate the strength of nodes to maintain stiff voltages [5], [6], [9], [19]. The VSF of node i is calculated as

$$\mathrm{VSF}_i = \lim_{\Delta Q_i \to 0} \frac{\Delta V_i}{\Delta Q_i} \tag{15}$$

High VSF means that even small changes in the reactive power causes large changes in the voltage magnitude and indicates that node i is weak.

Here, we derive the reactive power sensitivity index (QSF) to evaluate the voltage support capability of nodes, which is calculated as

$$\mathrm{QSF}_i = \lim_{\Delta U_i \to 0} -\frac{\Delta Q_i}{\Delta U_i} \tag{16}$$

High QSF means the node provides large reactive power to regulate the voltage, thus indicating strong capability to support voltage. According to equation (8) and (11), the QSF of SG and CVRPG nodes can be derived as

$$\mathrm{QSF}_{\mathrm{SG}} = \frac{2U_i - E'_i \cos\theta_{\delta i}}{X'_{di}} = \frac{U_i}{X'_{di}} - \frac{Q_{Gi}}{U_i} \tag{17}$$

$$\mathrm{QSF}_{\mathrm{RPG}} = \begin{cases} K_{Pi}U_i - \frac{Q_{Gi}}{U_i}, & i_{qi} < i_{q\max i} \\ 0, & i_{qi} \geq i_{q\max i} \end{cases} \tag{18}$$

By comparing equations (17) and (18), it can be seen that the voltage support capability of a SG node is mainly determined by the internal impedance, while that of a CVRPG nodes mainly depends on the proportional coefficient K_{Pi}. In addition, the equivalent internal impedance of CVRPG is the reciprocal of K_{P_i}. Both of them slightly decreases with the increase of the reactive power output. When the current of a CVRPG node reaches the limit, its voltage support capability decreases rapidly.

IV. Simulation Verification

The effectiveness of the proposed method is verified based on a modified 10-machine New England power system, as shown in Fig. 3. The original synchronous generators at buses 30, 31, 33, 34, and 36 are replaced with wind turbines that operate in the constant AC voltage control mode. The transient reactances of all synchronous generators are set to be 0.03 p.u.

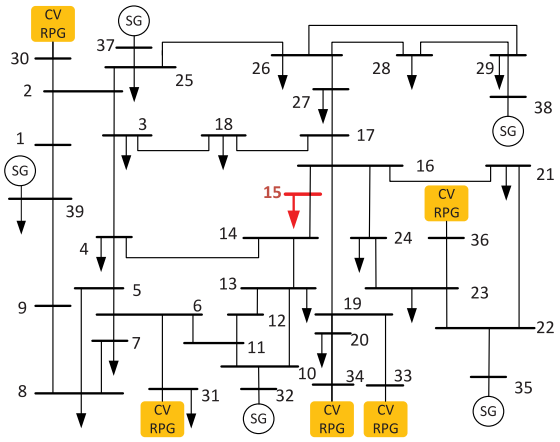


Fig. 3. Test power system: modified New England power system.

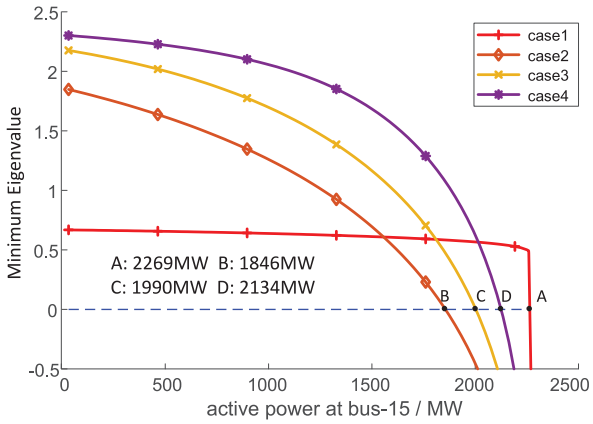


Fig. 4. Static voltage stability limit obtained by continuous current method.

A. Comparison of SVSA Results

The critical load power at Bus-15 is selected as the object of static voltage stability analysis, that is, the load power at Bus-15 is increased until the static stability limit is reached. To compare the SVSA accuracy of conventional CPF and extended CPF and to examine the impact of control parameters, the following four cases are concerned:

- *Case 1:* SVSA based on conventional CPF where CVRPG and SG nodes are treated as PV nodes;
- *Case 2:* SVSA based on extended CPF and $K_{Pi} = 0.5$;
- *Case 3:* SVSA based on extended CPF and $K_{Pi} = 1.0$;
- *Case 4:* SVSA based on extended CPF and $K_{Pi} = 2.0$.

The calculation results are displayed in Fig. 4, where each curve shows the minimum eigenvalues of the Jacobian matrix at different loading levels of Bus-15. A, B, C, and D points mark the static voltage stability limits of the four cases, which are 2269MW, 1846MW, 1990MW, and 2134MW.

Time-domain simulations in PSCAD provide verifications, as shown in Fig. 5. It can be seen that the critical power at Bus-15 in Case 3 is 1980MW, exceeding which the voltage collapses. Similarly, the power limit at Bus-15 in Case 4 is 2130 MW. They show high consistency with the results of the extended CPF proposed in this brief. While, 1980MW and 2130 MW are both indicated static voltage instability with conventional CPF. Thus, we can draw the conclusion that the

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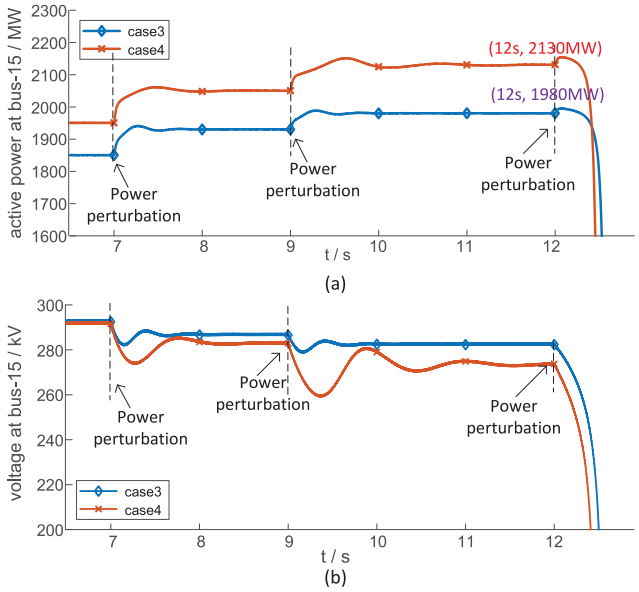


Fig. 5. Time-domain simulation results with PSCAD. (a) Load power at bus-15. (b) Voltage at bus-15.

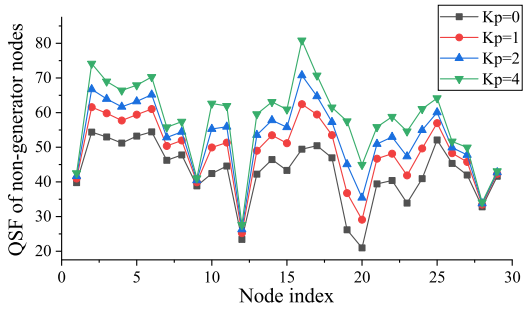


Fig. 6. Voltage support capability of nodes for different values of K_P.

extended CPF is more accurate than the conventional CPF in SVSA. By comparing B, C, and D points, we can also draw the conclusion that higher K_{Pi} in CVRPG nodes enhances the static voltage stability. This is due to the fact that a higher K_{Pi} makes the CVRPG have a stronger ability to response and suppress system voltage variations, i.e., to provide more reactive power for the same small disturbance.

B. Comparison of Voltage Support Capability

In order to more explicitly demonstrate the influence of the control parameters of CVRPG nodes on the voltage support capabilities, QSF of each node is calculated for different values of K_P as shown in Fig. 6. It can be seen that the voltage support capability of each node rises as K_P is increased.

V. Conclusion

In this brief, we propose an extended CPF method for SVSA of RPG-penetrated power systems, where an extended Jacobian matrix is established by considering the dynamics of SG and CVRPG nodes. Simulation results demonstrate that the proposed method has higher accuracy than conventional CPF and that the control parameters of CVRPG affect on the voltage stability of the system.

Furthermore, the reactive power sensitivity index is proposed to indicate the voltage support capability of nodes. The work can be used for figuring out the good operation conditions of power systems and optimizing control parameters of CVRPG, and will be further studied in our future work.

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